

1. Administrative & Study Identification

Full Study Title: A Study of Nivolumab, Nivolumab Plus Ipilimumab, or Investigator's Choice Chemotherapy for the Treatment of Participants With Deficient Mismatch Repair (dMMR)/Microsatellite Instability High (MSI-H) Metastatic Colorectal Cancer (mCRC) **Short Title/Acronym:** CheckMate 8HW (CheckMate -8HW) **Protocol Number:** CA209-8HW **NCT Number:** NCT04008030 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** Bristol Myers Squibb (BMS) **Investigational Product:** Nivolumab (Opdivo) and Ipilimumab (Yervoy) vs. Investigator's Choice Chemotherapy **Phase of Development:** Phase III **Medical Monitor:** Bristol Myers Squibb Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the clinical benefit, as measured by Progression-Free Survival (PFS), of nivolumab plus ipilimumab versus investigator's choice of chemotherapy in the first-line setting, and nivolumab plus ipilimumab versus nivolumab monotherapy across all lines of therapy. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Overall Survival (OS), Progression-Free Survival 2 (PFS2), and the safety/tolerability of the regimens.

2.2 Background & Rationale To address the critical need for more effective, durable treatments for patients with microsatellite instability-high (MSI-H) or mismatch repair-deficient (dMMR) metastatic colorectal cancer (mCRC). Standard chemotherapy often yields limited long-term benefits in this population. This study investigates whether dual immunotherapy (PD-1 and CTLA-4 inhibition) provides superior, durable progression-free survival and overall response compared to standard-of-care chemotherapy or anti-PD-1 monotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Nivolumab monotherapy or Investigator's Choice Chemotherapy) **Blinding:** Open-label **Randomization:** 2:2:1 (Nivolumab + Ipilimumab : Nivolumab Monotherapy : Chemotherapy)

3.2 Planned Enrollment & Duration Target \$N\$: 839 randomly assigned patients **Number of Sites:** Multinational, multicenter **Estimated Study Start:** 16-Aug-2019 **Estimated Primary Completion:** 10-Apr-2023 (Initial Data Cutoff: Aug-2024)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years (with pediatric provisions $\geq$ 12 years per FDA guidelines). 2. Histologically confirmed recurrent or metastatic colorectal cancer (CRC) not amenable to surgery. * *For Part 1 enrollment:* Irrespective of prior treatment history with chemotherapy and/or targeted agents. * *For Part 2 enrollment:* No prior treatment history with chemotherapy and/or targeted agents for metastatic disease. 3. Known tumor deficient mismatch repair (dMMR) or microsatellite instability-high (MSI-H) status per local standard of practice / centrally confirmed. 4. Eastern cooperative oncology group (ECOG) performance status lower than or equal to 1.

4.2 Exclusion Criteria 1. An active, known, or suspected autoimmune disease. 2. History of interstitial lung disease or pneumonitis. 3. Known history of positive test for human immunodeficiency virus (HIV) or known acquired immunodeficiency syndrome (AIDS).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Combination Arm:** Nivolumab 240 mg + Ipilimumab 1 mg/kg every 3 weeks for 4 doses, followed by Nivolumab 480 mg every 4 weeks. * **Monotherapy Arm:** Nivolumab 240 mg every 2 weeks for 6 doses, then 480 mg every 4 weeks. * **Chemotherapy Arm:** Investigator's choice (mFOLFOX-6 or FOLFIRI, which include fluorouracil [ATC Code: L01BC02; Trade names: Carac, Efudex, Adrucil, Fluoroplex, Tolak], with or without bevacizumab or cetuximab). **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Up to 2 years, or until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom management medications. **Prohibited:** Concurrent administration of other experimental anti-cancer therapies or immunosuppressive doses of systemic corticosteroids (unless managing an immune-mediated adverse event).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Tumor biopsy for central dMMR/MSI-H confirmation, Baseline Imaging
2	0	Randomization, First Dose of Study Treatment
3	3 to 4 Weeks	Treatment administration, Safety Labs, Efficacy/Tumor Assessments (typically every 8-12 weeks)
4	Up to 2 Years	Final Vital Signs, Safety Follow-up, Ongoing Survival and Progression Monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) comparing the combination arm vs. chemotherapy (1L setting) and combination arm vs. monotherapy (all lines). **Measurement Method:** Blinded Independent Central Review (BICR) using RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) by BICR - 1L Participants and All Randomized Participants. * Objective Response Rate (ORR) evaluated by BICR. * Overall Survival (OS). * Progression-Free Survival 2 (PFS2).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each visit, with a specific focus on identifying and managing immune-mediated adverse events (imAEs) related to dual checkpoint blockade. **Serious Adverse Events (SAEs):** Reported to the sponsor and regulatory authorities typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee for routine safety and efficacy oversight.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** Powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect a statistically significant improvement in PFS with the dual combination versus the control arms using log-rank testing. **Handling of Missing Data:** Censoring rules for time-to-event endpoints applied per RECIST v1.1 regulatory guidance.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical monitoring of participating sites to ensure protocol adherence and source data verification. **Audit Trail:** Secure Electronic Data Capture (EDC) system with strict eCRF locking protocols and query resolution workflows.

11. Study Identifiers & Governance

Primary ID: CA209-8HW **Posting ID:** 134361921438674349 **Registry Number:** NCT04008030 **Sponsor/Lead Organization:** Bristol Myers

Squibb **Study Title:** CheckMate 8HW **Official Regulatory Title:** A Phase 3 Randomized Clinical Trial of Nivolumab Alone, Nivolumab in Combination With Ipilimumab, or Investigator's Choice Chemotherapy in Participants With Microsatellite Instability High (MSI-H) or Mismatch Repair Deficient (dMMR) Metastatic Colorectal Cancer

12. Clinical Framework (Oncology/mCRC Specific)

Primary Condition: Metastatic Colorectal Cancer (mCRC) **Preferred Name:** Metastasis from malignant neoplasm of colon and/or rectum **Preferred UMLS Name:** Microsatellite Instability High Colorectal Cancer **Diagnosis Threshold:** Deficient Mismatch Repair (dMMR) / Microsatellite Instability High (MSI-H) **Medical Methodology:** Dual Immune Checkpoint Inhibitor Therapy (Anti-PD-1 + Anti-CTLA-4) **Optimization Target:** Maximizing Progression-Free Survival (PFS) and Depth of Response (ORR)

13. Study Procedures & Methodology

Management Pathway: First-line and subsequent-line oncology treatment algorithms. **Education Protocol (HCP):** Site initiation visits and investigator training on identifying and managing immune-mediated adverse events (imAEs). **Education Protocol (Patient):** Provision of patient alert cards detailing symptoms of immune-related toxicities. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) to ensure unbiased objective radiological tumor assessment.

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years of active treatment, followed by long-term survival follow-up. **Onsite Visit Cadence:** Every 3 weeks during induction; Every 4 weeks during maintenance phase. **Remote Monitoring Frequency:** Continuous survival follow-up post-treatment (e.g., via phone/records every 12 weeks). **Checklist Review Interval:** Radiological tumor assessments performed every 8 to 12 weeks depending on the study phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				